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Order-Jet Filters for Polylogarithmic Transseries

Sharp cancellation costs, Stokes-neutral curves, and a Stieltjes–digamma inverse

Prepared for Vladimir Reshetnikov and the ProveIt project

Abstract

We study finite differential operators in the order of $F_s(x) = -\mathrm{Li}_s(-\mathrm{e}^x)$, retaining both the Stokes jump and the exponentially small correction to the arithmetic lateral average. The classical completion formula turns an order polynomial Π into its values at $-\log n \pm \mathrm{i}\pi$. This gives a sharp cancellation theorem: the real polynomials cancelling both data at the first N actions form exactly the ideal generated by $\prod_{n=1}^{N}((u+\log n)^2+\pi^2)$. We also determine the separate optimal costs, including their change at integer and half-integer orders, and use Gelfond–Schneider to prove a nonvanishing rule for every pure integer-order jet. For each $N \geq 1$ a normalized degree-$N+1$ filter admits a unique convergent moving-order curve on which the entire jump is zero. Its detuning is explicitly asymptotic to a constant times e^{-Nx}, although its physical completion still starts at e^{-x}. At order zero, Jonquière's formula gives all jets in generalized Stieltjes constants. A positive endpoint-subtracted integral shows that $H(x) = \partial_s F_s(x)|_{s=0}$ is a real-analytic increasing bijection $\mathbb{R} \to (0,\infty)$. Its digamma core has an exact ordinary Borel representation. We prove convergent inverse charts at both endpoints and compute a model-specific arithmetic-averaging defect: the inverse-average error is asymptotic to $\pi^2 w^2 \mathrm{e}^{-2w}$, whereas the physical correction to the exact core inverse is only $(\log 2) w \mathrm{e}^{-2w}$. Full proofs, finite exact regressions, and independent high-precision integral checks are supplied.

Status. Conventional mathematical proofs; not externally refereed or Lean-formalized. The cancellation classification, moving-order curve, and assembled order-zero inverse theory are proposed contributions. Worldwide priority is not established. Classical completion, Gamma/Bell calculus, and general nonlinear inverse transport are explicitly credited.

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1 The research question and its source boundary

1.1 Three objects that must be kept separate

A terminating or factorially divergent power–logarithmic expansion does not determine a unique analytic function. In this report the distinction can be made exact. For $x > 0$ set

$$F_s(x) = -\operatorname{Li}_s(-\mathrm{e}^x), \qquad T_s(x) = F_s(-x) = \sum_{n\geq 1} \frac{(-1)^{n-1}\mathrm{e}^{-nx}}{n^s}. \tag{1.1}$$

The subscript is the *polylogarithm order*; the usual normalized Fermi–Dirac index is $s - 1$ [1]. There are two distinguished lateral realizations $L_s^{\pm}$ of the Sommerfeld expansion, satisfying

$$L_s^{\pm} = F_s + \mathrm{e}^{\mp \mathrm{i}\pi s} T_s. \tag{1.2}$$

Consequently the jump $L_s^+ - L_s^-$ and the physical correction $(L_s^+ + L_s^-)/2 - F_s$ are different data. Neither should be silently substituted for the other. We use *arithmetic average*, not "median," for $(L_s^+ + L_s^-)/2$.

For a real polynomial Π, applying $\Pi(\partial_s)$ changes the coefficient of the nth exponential into the value of Π at $-\log n \mp \mathrm{i}\pi$. We ask how many initial actions can be removed by an order jet of prescribed degree, how this cancellation behaves under small order detuning, and what analytic inversion does to the remaining completion. Here an *action* n simply means the exponential e^{-nx}; this terminology does not add a summability hypothesis.

1.2 What the repository already proves

The comparison snapshot is

`8f5afd4d7565df808359f6a0c22f87f79a119bb9`.

At that snapshot the package *Resonant Polylogarithm Transseries* [8], dated 11 October 2026, already proves an explicit lattice Borel formula, the subtracted Gamma kernel for positive-integer first order derivatives, and harmonic Bell-polynomial formulas for all late coefficients. It also proves sharp dilogarithm inverse-radius results. We use its notation and its completion theorem; we do not present those results as new. Its Research Question 4 concerns global higher-jet Borel normal forms, and Question 6 concerns periodic-weight cancellation. Our order-polynomial filters are a different, precisely specified cancellation mechanism, not a claimed solution of all the periodic-weight or higher-kernel questions.

The general nonlinear inverse-transport theorem and the obstruction to commuting arithmetic averaging with inversion already appear in *Nonlinear Stokes Transport under Logarithmic Inversion* [9]. The present inverse calculations specialize that architecture to an exact digamma core. Complete Fermi–Dirac expansions, including exponentially small terms, are classical; a directly inspected account is Gil–Segura–Temme, Section 3 and Lemma 3.1 [6]. Jonquière inversion and the Gamma and digamma identities are also classical [1, 2, 3].

The new assertions proposed here are the sharp jet-degree classification, its moving-order zero curves and nonvanishing consequences, and the specific endpoint/inverse theory of H. Several ingredients are elementary once the correct exact completion is chosen. Their value lies in a complete classification with sharp converses and analytic realizations, not in a claim that polynomial interpolation itself is new.

1.3 What was not inspected

The incoming directory listing at the pinned snapshot contained `ProveIt_Lambert_Polylog_Order_Jets_2026-10-12.zip`, `ProveIt_Quadratic_Polylog_Boundaries_2026-10-12.zip`, and `Resonant_Lambert_Inverse_Transseries_2026-10-12.zip`. Their binary contents could not be retrieved through the available text-only GitHub read route; alternative retrieval attempts did not supply them. Their names are *not* evidence of their theorem contents. Thus nonduplication relative to those

three archives, and relative to all unread canonical chapters, is not certified. The source audit records the actual read ranges and this limitation. No remote repository file was changed, and no new mathematical error was established in the accepted statements actually inspected.

2 Exact lateral completion and analytic continuation

2.1 The analytic realization used throughout

We first state the existing completion in the exact normalization needed below, and give its derivation to fix signs and domains.

Proposition 2.1 (Classical completion in the repository normalization)**.** *For $x > 0$ and $\operatorname{Re} s > 0$, the formal Sommerfeld series is*

$$\widehat{P}_s(x) = \frac{x^s}{\Gamma(s+1)} + 2\sum_{k\geq 1} \frac{\eta(2k)}{\Gamma(s-2k+1)} x^{s-2k}, \qquad \eta(z) = (1-2^{1-z})\zeta(z). \tag{2.1}$$

In the Borel convention $\mathcal{B}(\sum_{j\geq 1} a_j x^{-j})(\xi) = \sum_{j\geq 1} a_j \xi^{j-1}/(j-1)!$, the normalized minor is

$$b_s(\xi) = s\sum_{n\geq 1} \frac{(-1)^{n-1}}{n} \left\{(1+\xi/n)^{s-1} - (1-\xi/n)^{s-1}\right\}. \tag{2.2}$$

The upper and lower lateral realizations

$$L_s^{\pm}(x) = \frac{x^s}{\Gamma(s+1)} \left(1 + \int_0^{\infty \mathrm{e}^{\pm \mathrm{i}0}} \mathrm{e}^{-x\xi} b_s(\xi)\, \mathrm{d}\xi\right) \tag{2.3}$$

satisfy (1.2)*. Equivalently, with $a_\pm = \frac{1}{2} \pm \frac{\mathrm{i}x}{2\pi}$,*

$$L_s^{\pm}(x) = -\frac{(2\pi)^s \mathrm{e}^{\mp \mathrm{i}\pi s/2}}{\Gamma(s)} \zeta(1-s, a_\pm). \tag{2.4}$$

The last formula has a removable singularity at $s = 0$ and is entire in s. It supplies the continuation of the lateral realization to every order. All finite order derivatives may be taken in the exact completion.

Proof. For $\operatorname{Re} s > 0$ the Fermi integral is

$$F_s(x) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{t^{s-1}}{1+\mathrm{e}^{t-x}}\, \mathrm{d}t.$$

Split at $t = x$ and put $u = |t-x|$. This gives

$$F_s(x) = \frac{x^s}{\Gamma(s+1)} + \frac{1}{\Gamma(s)} \int_0^\infty \frac{(x+u)^{s-1} - \mathbf{1}_{u<x}(x-u)^{s-1}}{\mathrm{e}^u + 1}\, \mathrm{d}u. \tag{2.5}$$

The binomial expansion of (2.2) at zero has coefficient $2\eta(2k)s^{\underline{2k}}/(2k-1)!$ at ξ^{2k-1}. This proves the formal Borel identity. The lattice sum converges normally on compact continued sheets away from $\pm 1, \pm 2, \ldots$: the bracket in its nth term is $O(n^{-1})$ and its prefactor contributes another n^{-1}. The same estimates, with logarithmic factors, hold for finitely many order derivatives on compact order sets.

On a ray strictly above or below the positive axis the kernel has at most polynomial–logarithmic growth. One can verify this by splitting the sum at $n \asymp |\xi|$: the larger n use the preceding difference estimate, and the finitely many smaller n use the power growth of their binomials. Thus the tilted Laplace integrals converge. On the boundary their local powers are integrable when $\operatorname{Re} s > 0$.

In the nth integral substitute $\xi = nu/x$. Summing $\sum_{n\geq 1}(-1)^{n-1}e^{-nu} = 1/(e^u+1)$ turns the integral part of (2.3) into

$$\frac{1}{\Gamma(s)}\int_0^\infty \frac{(x+u)^{s-1} - (x-u)_\mp^{s-1}}{e^u+1}\,du.$$

The upper path gives argument $-\pi$ to $x-u$ when $u > x$. Subtracting (2.5) therefore gives

$$-\frac{e^{\mp i\pi(s-1)}}{\Gamma(s)}\int_x^\infty \frac{(u-x)^{s-1}}{e^u+1}\,du = e^{\mp i\pi s}T_s(x).$$

Endpoint domination by $v^{\epsilon-1}(1+|\log v|^r)$ on compact sets $\operatorname{Re} s \geq \epsilon > 0$, and exponential domination at infinity, justify the differentiations and interchanges. Finite Taylor expansion of the Borel germ with a remainder gives the asymptotic interpretation of (2.1); no infinite divergent sum was integrated term by term.

Jonquière's identity [1, Eq. 25.12.13], at $z = -e^x$, gives the lower formula in (2.4); conjugating for real s and then using analytic continuation gives the upper one. For $a_\pm$ in the right half-plane, $\zeta(1-s, a_\pm)$ has only its simple pole at $s = 0$, cancelled by the simple zero of $1/\Gamma(s)$. The small-argument series T_s is entire in s, normally on compact sets. Hence both the completion and the physical F_s continue entirely in the order, and finite order derivatives remain valid. □

Remark 2.2 (Continuation is not an unregularized boundary integral). For $\operatorname{Re} s \leq 0$, (2.4) is the definition of the continued lateral realization used here. We do not assert that the boundary powers in (2.3) become ordinary locally integrable functions there. In Section 6 we give a separate, ordinary meromorphic Borel representation for the first jet at zero. Differentiating the *whole* normalized expression matters: $\log x$ is not treated as a constant coefficient in an unjustified ordinary Borel transform.

2.2 A polynomial dictionary for the exponential data

For real σ and $\Pi \in \mathbb{R}[u]$, write

$$\begin{aligned}
\mathcal{F}_{\sigma,\Pi}(x) &= \Pi(\partial_s)F_s(x)|_{s=\sigma}\,,\\
\mathcal{L}^\pm_{\sigma,\Pi}(x) &= \Pi(\partial_s)L^\pm_s(x)\big|_{s=\sigma}\,,\\
C_{\sigma,\Pi} &= \tfrac{1}{2}(\mathcal{L}^+_{\sigma,\Pi} + \mathcal{L}^-_{\sigma,\Pi}) - \mathcal{F}_{\sigma,\Pi},\\
J_{\sigma,\Pi} &= \mathcal{L}^+_{\sigma,\Pi} - \mathcal{L}^-_{\sigma,\Pi}.
\end{aligned} \tag{2.6}$$

Put $D = \mathrm{d}/\mathrm{d}u$. The operators

$$R_\sigma = \cos(\pi\sigma + \pi D), \qquad I_\sigma = \sin(\pi\sigma + \pi D) \tag{2.7}$$

are their Taylor series in the nilpotent operator D on any fixed-degree polynomial space; all sums terminate.

Theorem 2.3 (Exact order-filter dictionary)**.** *Let $u_n = -\log n$ and $q = e^{-x}$. Then*

$$\mathcal{L}^\pm_{\sigma,\Pi} - \mathcal{F}_{\sigma,\Pi} = e^{\mp i\pi\sigma}\sum_{n\geq 1}\frac{(-1)^{n-1}}{n^\sigma}\Pi(u_n \mp i\pi)q^n, \tag{2.8}$$

$$C_{\sigma,\Pi} = \sum_{n\geq 1}\frac{(-1)^{n-1}}{n^\sigma}(R_\sigma\Pi)(u_n)q^n, \tag{2.9}$$

$$J_{\sigma,\Pi} = -2i\sum_{n\geq 1}\frac{(-1)^{n-1}}{n^\sigma}(I_\sigma\Pi)(u_n)q^n. \tag{2.10}$$

The series converge absolutely for $|q| < 1$, normally on compact order sets. For real u,

$$(R_\sigma\Pi)(u) + i(I_\sigma\Pi)(u) = e^{i\pi\sigma}\Pi(u + i\pi). \tag{2.11}$$

Proof. Every derivative of $e^{\mp i\pi s}n^{-s}$ multiplies it by $-\log n \mp i\pi$. Apply the finite operator to (1.2). A polynomial in $\log n$ times $n^{-\sigma}$ has subexponential growth in n, which proves the stated normal convergence. Taylor's formula for a polynomial gives $\Pi(u + i\pi) = \exp(i\pi D)\Pi(u)$ and hence (2.11). Taking the arithmetic average and difference proves the two remaining formulas, with the sign in the jump fixed by upper minus lower. □

3 Sharp cancellation costs and a complete joint ideal

3.1 What a degree can and cannot remove

For a nonzero power series in q with no constant term, let ν be its first nonzero exponent; let $\nu(0) = \infty$.

Theorem 3.1 (Sharp separate degree costs)**.** *Fix $\Pi \neq 0$ of degree d and real σ. If $\sin \pi\sigma \neq 0$, then $\nu(J_{\sigma,\Pi}) \leq d + 1$; if $\sin \pi\sigma = 0$ and $d \geq 1$, then $\nu(J_{\sigma,\Pi}) \leq d$. Similarly, if $\cos \pi\sigma \neq 0$, then $\nu(C_{\sigma,\Pi}) \leq d + 1$; if $\cos \pi\sigma = 0$ and $d \geq 1$, then $\nu(C_{\sigma,\Pi}) \leq d$. Every displayed bound is attained by a polynomial of the indicated degree. The exceptional constant polynomials have zero jump at integer order, and zero arithmetic completion at half-integer order.*

Proof. The constant term of I_σ as a polynomial in D is $\sin \pi\sigma$. When this is nonzero, I_σ is an invertible triangular operator on degree-at-most-d polynomials and preserves the degree. When it is zero,

$$I_\sigma = (\pi \cos \pi\sigma)D + O(D^3),$$

so it has kernel exactly the constants and reduces every positive degree by exactly one. The analogous assertions for R_σ follow from its constant and linear terms, $\cos \pi\sigma$ and $-\pi \sin \pi\sigma$.

A nonzero polynomial of degree k cannot vanish at all of the $k + 1$ distinct real numbers $u_1, \ldots, u_{k+1}$. The coefficients multiplying these evaluations in (2.9)–(2.10) never vanish. This proves the upper bounds.

To attain a bound for a degree-preserving operator, take its inverse image of $\prod_{n=1}^{d}(u + \log n)$. For a degree-lowering operator, its map from polynomials of degree at most d, modulo constants, to polynomials of degree at most $d - 1$ is an isomorphism. Take the inverse image of $\prod_{n=1}^{d-1}(u + \log n)$. The next node is not a root, so the bound is attained exactly. The assertions for constants follow directly from (2.7). □

Thus, for $d \geq 1$, the optimal first surviving actions are

Order center	Jump	Arithmetic completion
$\sigma \in \mathbb{Z}$	d	$d + 1$
$\sigma \in \mathbb{Z} + \frac{1}{2}$	$d + 1$	d
Neither of the above	$d + 1$	$d + 1$

The two columns are separately optimized; they need not be realized by the same polynomial. In particular, a degree statement about one datum cannot be reused for their simultaneous cancellation.

For example, at integer order the operators are, up to $(-1)^\sigma$, $\cos(\pi D)$ and $\sin(\pi D)$. A polynomial that delays the completion to action $d + 1$ is

$$\Pi_d^C(u) = \sec(\pi D) \prod_{n=1}^{d}(u + \log n). \tag{3.1}$$

The power series for sec terminates on the polynomial. A primitive followed by the finite operator $\pi D/\sin(\pi D)$ similarly constructs a jump extremizer. These formulas are exact algebra, not asymptotic infinite operator expansions.

3.2 Simultaneous cancellation is a principal ideal

Theorem 3.2 (Joint cancellation ideal and optimal real jet degree)**.** *Let E be any finite set of positive integers. For every real order center σ, the real polynomials Π for which both the jump and the arithmetic completion vanish at every action in E form exactly*

$$\left(\prod_{n\in E}\big((u+\log n)^2+\pi^2\big)\right)\mathbb{R}[u]. \tag{3.2}$$

Consequently, for $E=\{1,\dots,N\}$ the minimum nonzero degree is $2N$, and the unique monic minimizer is

$$Q_N^{\mathrm{joint}}(u)=\prod_{n=1}^{N}\big((u+\log n)^2+\pi^2\big). \tag{3.3}$$

At action $N+1$, at least one of its jump and completion coefficients is nonzero. More generally a degree-d nonzero jet cannot make both data zero at all of the first $\lfloor d/2\rfloor+1$ actions.

Proof. By (2.11), both real components at the node u_n vanish if and only if $\Pi(u_n+\mathrm{i}\pi)=0$. Since Π is real, this is equivalent to zeros at both $u_n+\mathrm{i}\pi$ and $u_n-\mathrm{i}\pi$, or to divisibility by $(u-u_n)^2+\pi^2$. These real quadratic factors are pairwise coprime: all their complex roots are distinct. Divisibility by each is therefore equivalent to divisibility by their product. This proves the ideal identity and the degree assertions.

For the monic minimizer at the next node, putting $\ell_j=\log(j/(N+1))$ gives

$$Q_N^{\mathrm{joint}}(-\log(N+1)+\mathrm{i}\pi)=\prod_{j=1}^{N}\ell_j(\ell_j+2\mathrm{i}\pi)\neq 0. \tag{3.4}$$

After rotation by $\mathrm{e}^{\mathrm{i}\pi\sigma}$ this still has at least one nonzero real component. This proves the final assertion. We do not claim that both components must separately be nonzero in this formula. □

Corollary 3.3 (First-order detuning robustness)**.** *For a fixed real polynomial Π and a finite action set E, the following requirements are equivalent: the jump coefficients at every $n\in E$ vanish together with their first order-center derivatives at one real center σ_0; both the jump and completion coefficients vanish there; and the full lateral completion coefficients vanish at those actions for* every *order center. Their common solution ideal is* (3.2)*. In particular robustness through first order in detuning costs exactly $2N$ for the first N actions.*

Proof. Up to a nonzero fixed scalar, the nth jump coefficient is $n^{-\sigma}(I_\sigma\Pi)(u_n)$. Its derivative is

$$n^{-\sigma}\{-\log n\,(I_\sigma\Pi)(u_n)+\pi(R_\sigma\Pi)(u_n)\}.$$

When the jump coefficient is zero, its derivative is zero exactly when the completion coefficient is zero. The joint theorem then forces $\Pi(u_n\pm\mathrm{i}\pi)=0$, independently of the order center. The converse is immediate from the exact dictionary. □

Corollary 3.4 (Finite reconstruction from exponential data)**.** *Fix N and real σ. A polynomial of degree less than $2N$ is uniquely determined by its jump and arithmetic-completion coefficients at the first N actions. Every assignment of those $2N$ real normalized data is attained by exactly one such polynomial.*

Proof. Divide out the nonzero scalar coefficients in (2.9)–(2.10) and undo the rotation in (2.11). This specifies Π at the N upper nodes and, by conjugacy, at the N lower nodes. Ordinary interpolation at $2N$ distinct complex points gives a unique degree-less-than-$2N$ polynomial. Its conjugate-reflection polynomial has the same values, so uniqueness makes its coefficients real. Equivalently, injectivity follows from theorem 3.2 and equal finite dimensions give surjectivity. □

The first joint filter. For $N = 1$, the operator is $\partial_s^2 + \pi^2$. It removes the entire first action, not just its imaginary part. At integer m the next completion coefficient is $(-1)^{m+1}(\log 2)^2/2^m$ and the next jump coefficient is $-4\pi\mathrm{i}(-1)^m \log 2/2^m$. This is a simple exact test of the real-versus-imaginary distinction.

3.3 An explicit finite-tail estimate

For numerical evaluation the exact coefficient dictionary also supplies an elementary bound. Let $\Pi(u) = \sum_{j=0}^{d} p_j u^j$ and $K_\Pi = \sum_{j=0}^{d} |p_j|(1+\pi)^j$. For real $\sigma \geq 0$, $0 < q < 1$, and integer $M \geq 0$, each full lateral completion tail obeys

$$\left| \sum_{n>M} \frac{(-1)^{n-1}}{n^\sigma} \Pi(-\log n \pm \mathrm{i}\pi) q^n \right| \leq K_\Pi (M+1)^{d-\sigma} q^{M+1} \frac{E_d(q)}{(1-q)^{d+1}}, \tag{3.5}$$

where E_d is the polynomial defined by $\sum_{k\geq 0}(k+1)^d q^k = E_d(q)/(1-q)^{d+1}$, with $E_0 = 1$. Indeed $|\Pi(-\log n \pm \mathrm{i}\pi)| \leq K_\Pi n^d$ for $n \geq 1$, $n = M + 1 + k \leq (M+1)(k+1)$, and $n^{-\sigma} \leq (M+1)^{-\sigma}$. This proves the bound. The jump tail has at most twice the right side; the arithmetic-completion tail has at most the right side. This is a finite evaluation safeguard, not the basis of the cancellation theorem.

4 Pure-jet rigidity and exact Stokes-neutral order curves

4.1 A transcendence input rules out accidental zeros

Lemma 4.1. *For every integer $n \geq 2$, $\log n/\pi$ is transcendental.*

Proof. Use the classical Gelfond–Schneider theorem in its complex form: if $\alpha \neq 0, 1$ and $\beta \notin \mathbb{Q}$ are algebraic, every fixed value of α^β is transcendental [7, Theorem 3.1]. If $a = \log n/\pi$ were algebraic, $\beta = -\mathrm{i}a$ would be nonreal algebraic, hence irrational. With $\log(-1) = \mathrm{i}\pi$, $(-1)^{-\mathrm{i}a} = \exp(\pi a) = n$, a contradiction. □

Theorem 4.2 (Parity and nonvanishing for every pure jet)**.** *Fix an integer order m and $r \geq 1$, and take $\Pi(u) = u^r$. At every action $n \geq 2$, both the jump coefficient and the arithmetic-completion coefficient are nonzero. Their first surviving actions are exactly*

$$(\nu(J_{m,u^r}), \nu(C_{m,u^r})) = \begin{cases} (1,2), & r \text{ odd}, \\ (2,1), & r \text{ even}. \end{cases} \tag{4.1}$$

For $r = 0$, the jump is identically zero and the completion starts at one. At half-integer centers the two columns of (4.1) are exchanged.

Proof. At $n = 1$, the relevant complex value is $(\mathrm{i}\pi)^r$; its real and imaginary parts alternate with the parity of r. At $n \geq 2$, divide $(-\log n + \mathrm{i}\pi)^r$ by π^r. Its real and imaginary parts are nonzero polynomials in $a = \log n/\pi$ with integer coefficients: the real polynomial has leading term $(-a)^r$, and the imaginary one has leading term $r(-a)^{r-1}$. Neither can vanish at the transcendental number in theorem 4.1. At integer order the rotation is just $(-1)^m$; at half-integer order it is $\pm\mathrm{i}$, which exchanges the two components. The constant case follows from the completion formula. □

Corollary 4.3 (Rigidity for algebraic normalized coefficients)**.** *The nonvanishing at every action $n \geq 2$ extends from pure monomials to all nonconstant filters $\Pi(u) = c\,p(u/\pi)$, where $c \in \mathbb{R} \setminus \{0\}$ and p has real algebraic coefficients. At integer or half-integer centers no such filter can cancel even one of the two data at action two. In particular arbitrarily high degree in this coefficient class cannot reproduce the deep logarithmic-node filters of theorem 3.1.*

Proof. The real and imaginary parts of $p(-a + \mathrm{i})$ are nonzero polynomials in a with real algebraic coefficients: their degrees are respectively $\deg p$ and $\deg p - 1$. A number transcendental over $\mathbb{Q}$ cannot be a root of a nonzero polynomial over the algebraic numbers. Apply theorem 4.1 and then the integer or half-integer rotation. The common nonzero real scale c does not affect vanishing. □

This use of Gelfond–Schneider proves an all-order nonvanishing statement; a finite table of numerical coefficients would not prove it. Conversely, the freely chosen real coefficients in theorem 3.2 deliberately permit cancellation. There is no contradiction: a pure monomial in the order derivative is a much smaller class of filters.

4.2 A normalized sharp jump filter

For $N \geq 1$ define

$$Q_N(u) = \prod_{j=1}^{N} (u + \log j). \tag{4.2}$$

There is a unique real polynomial Π_N with

$$\sin(\pi D)\Pi_N = Q_N, \qquad (\cos(\pi D)\Pi_N)(0) = 1. \tag{4.3}$$

It has degree $N + 1$. Existence and uniqueness follow from the isomorphism modulo constants used in theorem 3.1; the second condition fixes that constant. One constructive formula is

$$\Pi_N = \frac{1}{\pi} \frac{\pi D}{\sin(\pi D)} \int_0^u Q_N(v)\, \mathrm{d}v + c_N,$$

where c_N enforces the normalization. All operators terminate. Writing $R_N = \cos(\pi D)\Pi_N$, the first two examples are

$$\Pi_1(u) = \frac{u^2}{2\pi} + 1 + \frac{\pi}{2}, \qquad R_1(u) = 1 + \frac{u^2}{2\pi}, \tag{4.4}$$

$$\Pi_2(u) = \frac{u^3}{3\pi} + \frac{\log 2}{2\pi} u^2 + \frac{\pi}{3} u + 1 + \frac{\pi \log 2}{2}, \qquad R_2(u) = 1 + \frac{u^3}{3\pi} + \frac{\log 2}{2\pi} u^2 - \frac{2\pi}{3} u. \tag{4.5}$$

The normalization is intentionally about the *completion*, not about $\Pi_N(0)$.

Theorem 4.4 (Convergent Stokes-neutral curve and its crossover)**.** *Fix an integer $m \geq 0$ and $N \geq 1$. For sufficiently small q, there is a unique holomorphic function $\epsilon_N(q)$ with $\epsilon_N(0) = 0$ for which*

$$J_{m+\epsilon_N(q),\Pi_N}(-\log q) = 0 \tag{4.6}$$

when $q > 0$. For complex q, this identity refers to the convergent q-series of the jump. The function has real Taylor coefficients and

$$\epsilon_N(q) = -\frac{\kappa_{m,N}}{\pi} q^N + O(q^{N+1}), \qquad \kappa_{m,N} = \frac{1}{(N+1)^m} \prod_{j=1}^{N} \log \frac{N+1}{j} > 0. \tag{4.7}$$

The associated completion does not *disappear:*

$$C_{m+\epsilon_N(q),\Pi_N}(-\log q) = (-1)^m q + O(q^2). \tag{4.8}$$

More generally, with $\epsilon = \lambda q^N$,

$$\frac{J_{m+\lambda q^N,\Pi_N}(-\log q)}{-2\mathrm{i}(-1)^m q^{N+1}} = \pi\lambda + \kappa_{m,N} + O(q), \tag{4.9}$$

uniformly for λ in any fixed compact subset of $\mathbb{C}$. At $m = 0$ these are analytically continued lateral realizations; no unregularized order-zero boundary integral is assumed.

Proof. Remove the factor $-2\mathrm{i}(-1)^m q$ from the jump and define

$$\Psi(\epsilon, q) = \sum_{n\geq 1} (-1)^{n-1} n^{-m-\epsilon} q^{n-1} \{\sin(\pi\epsilon) R_N(u_n) + \cos(\pi\epsilon) Q_N(u_n)\}. \tag{4.10}$$

The series is holomorphic for $|q| < 1$ and ϵ in a neighborhood of zero: on compact sets its coefficients have at most a fixed power of n times a fixed power of $\log n$. Its $n = 1$ term is precisely $\sin(\pi\epsilon)$, since $Q_N(0) = 0$ and $R_N(0) = 1$. Consequently

$$\Psi(0, 0) = 0, \qquad \partial_\epsilon \Psi(0, 0) = \pi.$$

The holomorphic implicit-function theorem gives the unique curve. Conjugation and uniqueness imply real Taylor coefficients.

At $\epsilon = 0$, the first N coefficients vanish. At $n = N + 1$,

$$(-1)^N (N + 1)^{-m} Q_N(-\log(N + 1)) = \kappa_{m,N}.$$

Thus $\Psi(0, q) = \kappa_{m,N} q^N + O(q^{N+1})$. Substitution in the implicit equation, whose ϵ derivative is $\pi + O(q)$, gives (4.7). Taylor expansion with $\epsilon = \lambda q^N$ proves (4.9); higher sine terms and mixed terms are of sufficiently high order, and holomorphy gives uniformity on compact λ sets.

The corresponding completion has normalized coefficient

$$\cos(\pi\epsilon) R_N(u_n) - \sin(\pi\epsilon) Q_N(u_n).$$

Its first term is $(-1)^m q \cos(\pi\epsilon)$. The rest is $O(q^2)$ uniformly for small ϵ, and the implicit curve tends to zero. This proves (4.8). □

Remark 4.5 (What the curve cancels). For a *fixed* order, the sharp filters remove individual initial action coefficients. On the moving-order curve, coefficients at different actions compensate after ϵ is substituted as a function of q. The result is an exact identity of analytic functions, not a statement that every fixed-order Borel singularity has disappeared. It also gives a precise example in which zero total lateral jump does not identify the physical function with the common lateral realization.

5 All order-zero jets in Stieltjes and integral coordinates

5.1 A complete finite formula in generalized Stieltjes constants

For $\mathrm{Re}\, a > 0$ use the Hurwitz–Stieltjes convention [5]

$$\zeta(1 + z, a) = \frac{1}{z} + \sum_{n\geq 0} \frac{(-1)^n \gamma_n(a)}{n!} z^n. \tag{5.1}$$

In particular $\gamma_0(a) = -\psi(a)$. Define the complete exponential Bell polynomials by

$$\exp\left(\sum_{j\geq 1} b_j \frac{z^j}{j!}\right) = \sum_{r\geq 0} \mathcal{B}_r(b_1, \ldots, b_r) \frac{z^r}{r!}, \qquad \mathcal{B}_0 = 1. \tag{5.2}$$

The following theorem specializes classical Jonquière inversion to a resonance excluded from the positive-integer Gamma-kernel statement of [8].

Theorem 5.1 (Stieltjes coordinates for every zero-order lateral jet)**.** *Let $x > 0$, $a_\pm = \frac{1}{2} \pm \frac{\mathrm{i}x}{2\pi}$, and*

$$\begin{aligned} d_1^\pm &= \gamma + \log(2\pi) \mp \frac{\mathrm{i}\pi}{2}, \\ d_j^\pm &= (-1)^{j+1} (j - 1)! \zeta(j), \qquad j \geq 2. \end{aligned} \tag{5.3}$$

For $r \geq 1$ put

$$Z_r^{\pm}(x) = \mathcal{B}_r(d^{\pm}) - r\sum_{n=0}^{r-1}\binom{r-1}{n}\gamma_n(a_{\pm})\mathcal{B}_{r-1-n}(d^{\pm}), \qquad Z_0^{\pm} = 1. \tag{5.4}$$

Then $Z_r^{\pm} = \partial_s^r L_s^{\pm}|_{s=0}$ and

$$\partial_s^r F_s(x)\big|_{s=0} = Z_r^{\pm}(x) - \sum_{n\geq 1}(-1)^{n-1}(-\log n \mp \mathrm{i}\pi)^r \mathrm{e}^{-nx}. \tag{5.5}$$

The formulas hold for every finite r and locally uniformly for $x > 0$.

Proof. The reciprocal-Gamma expansion [2, §5.7] gives

$$\frac{1}{\Gamma(s)} = s\exp\left(\gamma s + \sum_{j\geq 2}\frac{(-1)^{j+1}\zeta(j)}{j}s^j\right), \qquad |s| < 1.$$

In (5.1) substitute $z = -s$. Combining it with (2.4) yields the exact local generating identity

$$L_s^{\pm}(x) = \exp\left(\sum_{j\geq 1} d_j^{\pm}\frac{s^j}{j!}\right)\left(1 - \sum_{n\geq 0}\gamma_n(a_{\pm})\frac{s^{n+1}}{n!}\right). \tag{5.6}$$

In particular the removable value at zero is 1, not zero. Multiplying Taylor series and taking $r!$ times the coefficient gives (5.4). Applying the exact completion formula at zero gives (5.5). The local Taylor identities and the normally convergent exponential series justify every finite coefficient extraction and the claimed local uniformity. □

For example,

$$Z_1^{\pm} = d_1^{\pm} - \gamma_0(a_{\pm}), \qquad Z_2^{\pm} = (d_1^{\pm})^2 - \zeta(2) - 2d_1^{\pm}\gamma_0(a_{\pm}) - 2\gamma_1(a_{\pm}).$$

The order-zero second physical derivative is therefore an explicit combination of γ_1 at a conjugate pair of points, digamma values, and a convergent log-weighted exponential series. There is no finite-part ambiguity in these identities: the regularization is fixed by the holomorphic product in (2.4).

5.2 Independent endpoint-subtracted integral coordinates

A second exact representation is useful both for proof and for independent numerical evaluation. It avoids differentiating a numerically delicate polylogarithm continuation at an integer order.

Theorem 5.2 (Positive regularization and all zero-order moments)**.** *For any real x let $q = \mathrm{e}^{-x}$ and $f_0 = (1+q)^{-1}$. Define*

$$M_j(x) = \frac{1}{1+q}\int_0^{\infty}\frac{(1-\mathrm{e}^{-t})(\log t)^j}{t(1+\mathrm{e}^{t-x})}\,\mathrm{d}t, \qquad j \geq 0. \tag{5.7}$$

All these integrals are absolutely convergent. With $g_1 = \gamma$ and $g_j = (-1)^{j+1}(j-1)!\zeta(j)$ for $j \geq 2$,

$$F_0(x) = f_0, \qquad \partial_s^r F_s(x)\big|_{s=0} = r\sum_{j=0}^{r-1}\binom{r-1}{j}\mathcal{B}_{r-1-j}(g)M_j(x) \quad (r \geq 1). \tag{5.8}$$

In particular $H(x) := \partial_s F_s(x)|_{s=0} = M_0(x) > 0$.

Proof. Let $f_x(t) = (1 + e^{t-x})^{-1}$. The subtraction has the exact positive form

$$f_x(t) - f_0 e^{-t} = \frac{1 - e^{-t}}{(1 + e^{-x})(1 + e^{t-x})}. \tag{5.9}$$

For $\operatorname{Re} s > 0$, subtract the elementary Gamma integral to get

$$F_s(x) = f_0 + \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} (f_x(t) - f_0 e^{-t})\, dt. \tag{5.10}$$

At zero the subtracted numerator is $O(t)$, so the remaining integral is holomorphic for $\operatorname{Re} s > -1$. At infinity it has exponential decay on compact real x sets. Multiplication by $1/\Gamma(s)$ continues (5.10) to a neighborhood of zero. Its jth integral derivative at zero is $M_j(x)$; the majorants $t^{-1+\epsilon}(1 + |\log t|^j)$ after subtraction near zero, with $\epsilon > 0$, and exponential majorants at infinity justify this. The reciprocal-Gamma Bell expansion used above now gives (5.8). Positivity of the integrand for $j = 0$ proves the final assertion. □

6 The order-zero digamma core and its exact Borel kernel

6.1 An exact physical identity

Put

$$A(x) = \gamma + \log(2\pi) + \operatorname{Re} \psi\left(\tfrac{1}{2} + \frac{ix}{2\pi}\right), \qquad x > 0, \tag{6.1}$$

$$B(z) = \sum_{n \geq 2} (-1)^n (\log n) z^n = (\log 2) z^2 - (\log 3) z^3 + (\log 4) z^4 - \cdots, \qquad |z| < 1. \tag{6.2}$$

The letter B here denotes a convergent analytic function, not the Borel transform $\mathcal{B}$.

Proposition 6.1 (Digamma core and both lateral first jets)**.** *For $x > 0$,*

$$H(x) = A(x) - B(e^{-x}), \tag{6.3}$$

$$Z_1^{\pm}(x) = A(x) \mp i\pi \frac{e^{-x}}{1 + e^{-x}}. \tag{6.4}$$

Thus the first zero-order jet has a jump beginning at action one, while its physical correction to the arithmetic average begins at action two.

Proof. By theorem 5.1, $Z_1^+ = \gamma + \log(2\pi) - i\pi/2 + \psi(a_+)$. The digamma reflection formula [2, §5.5] gives

$$\psi(1 - z) - \psi(z) = \pi \cot(\pi z), \qquad \operatorname{Im} \psi\left(\tfrac{1}{2} + \frac{ix}{2\pi}\right) = \frac{\pi}{2} \tanh(x/2).$$

Hence the imaginary part of Z_1^+ is $-\pi/(e^x + 1)$, and its real part is A. The lower formula follows by conjugation. In (5.5) with $r = 1$, the imaginary tail cancels exactly, while the real tail is $B(e^{-x})$. □

Corollary 6.2 (Symmetry and the global positive core branch)**.** *The even function A satisfies*

$$H(x) + H(-x) = A(x), \qquad A(0) = \log(\pi/2), \qquad H(0) = \tfrac{1}{2} \log(\pi/2).$$

It is strictly increasing for $x > 0$ and maps that half-line onto $(\log(\pi/2), \infty)$.

Proof. For $x > 0$, $H(-x) = B(e^{-x})$, giving the symmetry from (6.3); continuity and evenness give the remaining real points. The classical value $\psi(1/2) = -\gamma - 2\log 2$ gives $A(0)$. For $v = x/(2\pi) > 0$, the convergent trigamma series gives

$$A'(x) = -\frac{1}{2\pi} \operatorname{Im} \sum_{k \geq 0} \frac{1}{(k + 1/2 + iv)^2} > 0,$$

because each summand has strictly negative imaginary part. The range then follows from the logarithmic asymptotic proved below. This also shows that the exact positive core inverse used later is unique wherever its target exceeds $\log(\pi/2)$. □

The single meromorphic expression, holomorphic near large real points, is

$$A(z) = \gamma + \log(2\pi) + \frac{1}{2}\left\{\psi\left(\tfrac{1}{2} + \frac{\mathrm{i}z}{2\pi}\right) + \psi\left(\tfrac{1}{2} - \frac{\mathrm{i}z}{2\pi}\right)\right\}. \tag{6.5}$$

It is meromorphic in z, with possible poles on the imaginary axis, and agrees with (6.1) on the positive real axis. In particular it is holomorphic on every disk of a fixed radius around a sufficiently large positive real w. We use this expression, not the nonholomorphic operation Re, in complex inverse equations.

6.2 A meromorphic minor at the previously excluded order

Theorem 6.3 (Ordinary Borel reconstruction at order zero)**.** *The formal algebraic tail of* $A(x) - (\log x + \gamma)$ *is*

$$A(x) \sim \log x + \gamma - 2\sum_{k\geq 1} \eta(2k)(2k-1)!\,x^{-2k}. \tag{6.6}$$

Its ordinary Borel minor is the meromorphic function

$$b_0^{\mathrm{jet}}(\xi) = \frac{1}{\xi} - \frac{\pi}{\sin \pi\xi}, \tag{6.7}$$

with a removable zero at the origin and residue $(-1)^{n-1}$ *at* $\xi = n \geq 1$*. For* $x > 0$*,*

$$\log x + \gamma + \int_{\pm} \mathrm{e}^{-x\xi} b_0^{\mathrm{jet}}(\xi)\,\mathrm{d}\xi = Z_1^{\pm}(x), \tag{6.8}$$

$$A(x) = \log x + \gamma + \mathrm{PV}\int_0^{\infty} \mathrm{e}^{-x\xi} b_0^{\mathrm{jet}}(\xi)\,\mathrm{d}\xi. \tag{6.9}$$

The principal value is the arithmetic average of the lateral integrals, or the sum of symmetric local principal values at the positive integers.

Proof. The classical generator $\pi\xi/\sin\pi\xi = 1 + 2\sum_{k\geq 1}\eta(2k)\xi^{2k}$ shows that

$$b_0^{\mathrm{jet}}(\xi) = -2\sum_{k\geq 1}\eta(2k)\xi^{2k-1}.$$

The ordinary Borel convention gives exactly the coefficients in (6.6). The residue statement follows from $(\sin\pi\xi)'|_{\xi=n} = \pi(-1)^n$.

For an independent proof of the exact reconstruction, initially take $\operatorname{Re} x > 0$ and $\operatorname{Im} x < 0$. Rotate an upper tilted Laplace path to $\xi = \mathrm{i}t$. There are no poles inside the rotation sector. On its large arcs, the cosecant has exponential decay away from the real axis and the $1/\xi$ term is suppressed by $\mathrm{e}^{-x\xi}$; the negative imaginary part of x ensures decay even at the final imaginary ray. The removable origin gives no small-circle contribution. Thus

$$\begin{aligned}\int_{+} \mathrm{e}^{-x\xi} b_0^{\mathrm{jet}}(\xi)\,\mathrm{d}\xi &= \int_0^{\infty} \mathrm{e}^{-\mathrm{i}xt}\left(\frac{1}{t} - \frac{\pi}{\sinh \pi t}\right)\mathrm{d}t \\ &= \int_0^{\infty}\left\{\frac{\mathrm{e}^{-\mathrm{i}xu/(2\pi)}}{u} - \frac{\mathrm{e}^{-(1/2+\mathrm{i}x/(2\pi))u}}{1-\mathrm{e}^{-u}}\right\}\mathrm{d}u.\end{aligned}$$

The standard digamma integral, combined with the Frullani difference, is [3, §5.9]

$$\psi(a) = \int_0^{\infty}\left(\frac{\mathrm{e}^{-u}}{u} - \frac{\mathrm{e}^{-au}}{1-\mathrm{e}^{-u}}\right)\mathrm{d}u, \quad \operatorname{Re} a > 0,$$

where cancellation at zero is taken inside the integrand. Consequently the preceding integral is

$$\psi\left(\tfrac{1}{2}+\frac{\mathrm{i}x}{2\pi}\right)-\log x+\log(2\pi)-\frac{\mathrm{i}\pi}{2}. \tag{6.10}$$

Both sides continue to a neighborhood of the positive real axis. A fixed upper ray of angle less than $\pi/2$ has an absolutely convergent Laplace integral there, so (6.10) proves the upper formula for real positive x. Conjugation proves the lower formula. Its value is precisely (6.4).

On the real boundary the only singularities are simple poles. Upper minus lower indentation contributes minus $2\pi\mathrm{i}$ times their residues, and the weighted residue sum is absolutely convergent:

$$-2\pi\mathrm{i}\sum_{n\geq 1}(-1)^{n-1}\mathrm{e}^{-nx}=-2\pi\mathrm{i}\frac{\mathrm{e}^{-x}}{1+\mathrm{e}^{-x}}.$$

At infinity one may take boundaries midway between the integer poles; exponential decay controls the remaining contour and principal-value tails. Averaging proves (6.9). Finally, a finite Taylor expansion of the germ on a tilted ray, with a Laplace remainder, gives (6.6) to each finite order. □

The kernel (6.7) is the natural $m = 0$ extension of the subtracted Gamma kernel in [8, label `thm:Gamma-kernel`]. The extension has been proved here with a separate contour calculation, rather than by asserting that all hypotheses of the $m \geq 1$ theorem persist unchanged.

Corollary 6.4 (Derivative scales for the exact core)**.** *As $w \to +\infty$,*

$$A(w)=\log w+\gamma-\frac{\pi^2}{6w^2}+O(w^{-4}), \tag{6.11}$$

$$A'(w)=\frac{1}{w}+O(w^{-3}),\qquad A''(w)=-\frac{1}{w^2}+O(w^{-4}). \tag{6.12}$$

The estimates are holomorphically uniform in a fixed closed subsector about the positive real axis, away from the origin. In particular, $A'(w) > 0$ for all sufficiently large real w.

Proof. Use the first term of (6.6). Uniformity follows from the tilted Laplace proof, or the standard sectorial digamma expansion [4, Eq. 5.11.2], applied to the conjugate-pair formula (6.5). Cauchy estimates on disks of radius a fixed positive multiple of w, contained in such a subsector, justify differentiation of each finite asymptotic remainder and give (6.12). This uses holomorphic remainder control, not mere differentiation of a real-axis big-O statement. □

7 A global inverse and convergent charts at both endpoints

7.1 The real inverse exists globally

Theorem 7.1 (Global order-zero inverse)**.** *The function $H(x) = \partial_s F_s(x)|_{s=0}$ is a real-analytic strictly increasing bijection from $\mathbb{R}$ to $(0,\infty)$. Its inverse $X(y)$ is real analytic on $(0,\infty)$. Its two endpoint scales are*

$$H(x)\sim(\log 2)\mathrm{e}^{2x}\quad(x\to-\infty),\qquad H(x)\sim\log x\quad(x\to+\infty). \tag{7.1}$$

Proof. Use the positive integral $H = M_0$ from theorem 5.2. Both factors $(1+\mathrm{e}^{-x})^{-1}$ and $(1+\mathrm{e}^{t-x})^{-1}$ increase strictly with x for every $t > 0$. Differentiation on compact x intervals is justified by the same endpoint and exponential majorants, and the integrand derivative is strictly positive. Hence $H'(x) > 0$. Local holomorphic continuation of the integral in x in a strip of sufficiently small width gives real analyticity.

For $x < 0$, the original small-argument polylogarithm series and normal order differentiation give $H(x) = B(\mathrm{e}^x)$, so $H(x) \sim (\log 2)\mathrm{e}^{2x} \to 0$. For large positive x, (6.3) and (6.11) give $H(x) = \log x + \gamma + O(x^{-2}) \to \infty$. Continuity and strict increase prove the range assertion, and the analytic inverse-function theorem applies everywhere because $H' > 0$. □

7.2 The small-target logarithmic–square-root chart

Write $a = \log 2$ and

$$b(z) = \frac{B(z)}{az^2} = 1 - \frac{\log 3}{a}z + \frac{\log 4}{a}z^2 - \cdots, \qquad b(0) = 1. \tag{7.2}$$

The square-root parameter for a small target is $t = (y/a)^{1/2} > 0$.

Theorem 7.2 (All coefficients and an explicit convergent endpoint disk)**.** *For $0 < t < \sqrt{3}/40$, the global inverse has the convergent expansion*

$$X(at^2) = \log t + \sum_{N\geq 1} c_N t^N, \qquad c_N = \frac{1}{N}[z^N]b(z)^{-N/2}. \tag{7.3}$$

The analytic correction $X(at^2) - \log t$ extends holomorphically to $|t| < \sqrt{3}/40$. In particular,

$$X(y) = \frac{1}{2}\log\frac{y}{a} + \frac{\log 3}{2a}\left(\frac{y}{a}\right)^{1/2} + \left(\frac{(\log 3)^2}{2a^2} - 1\right)\frac{y}{a} + O(y^{3/2}). \tag{7.4}$$

The radius given here is a sufficient certificate, not an exact radius.

Proof. Let $r = 1/20$. Since $\log n \leq n$ for $n \geq 1$,

$$\sup_{|z|\leq r}|b(z) - 1| \leq \frac{1}{a}\sum_{n\geq 3} nr^{n-2} = \frac{r(3-2r)}{a(1-r)^2} = \frac{58}{361a} < \frac{1}{4}.$$

For the strict last inequality, $\log 2 > 2/3$ follows, for example, from $\log 2 = 2\sum_{k\geq 0}(1/3)^{2k+1}/(2k+1)$; then $58/(361a) < 87/361 < 1/4$. There is a holomorphic square root of b on the disk, equal to one at zero, and $|\sqrt{b(z)}| > \sqrt{3}/2$ there. The function $f(z) = z\sqrt{b(z)}$ has exactly the simple zero 0, and on $|z| = r$ its modulus exceeds $\sqrt{3}/40$.

For $|t| < \sqrt{3}/40$, Rouché's theorem gives exactly one root, counting multiplicity, of $f(z) - t$ inside $|z| < r$. That root is simple, so it defines a holomorphic function $z(t)$ with $z(0) = 0$ and $z'(0) = 1$. Uniqueness glues the local roots throughout the disk. The quotient $z(t)/t$ extends holomorphically and has no zeros; its logarithm with value zero at $t = 0$ therefore exists on the disk. For positive small t, the root is real and positive and $H(\log z(t)) = B(z(t)) = at^2$. The global uniqueness theorem identifies $\log z(t)$ with $X(at^2)$.

For the coefficient formula, set $\Phi(z) = b(z)^{-1/2}$. The equation is $z = t\Phi(z)$ and $\log(z/t) = \log\Phi(z)$. Lagrange–Bürmann inversion gives, for $N \geq 1$,

$$\begin{aligned}[t^N]\log(z(t)/t) &= \frac{1}{N}[z^{N-1}](\log\Phi)'\Phi^N \\ &= \frac{1}{N^2}[z^{N-1}](\Phi^N)' = \frac{1}{N}[z^N]\Phi^N.\end{aligned}$$

This proves (7.3). Expanding the first two coefficients and using $\log 4 = 2\log 2$ proves (7.4). □

7.3 The large-target exact-core chart

For large $w > 0$, take the target $y = A(w)$, so w is the exact inverse of the algebraic digamma core rather than an approximation to that inverse. Put

$$q = \mathrm{e}^{-w}, \qquad a_w = A'(w), \qquad b_w = A''(w), \qquad c_+ = -\mathrm{i}\pi, \quad c_- = \mathrm{i}\pi. \tag{7.5}$$

Let $G_\pm(y)$ denote the inverse branches near w of the lateral functions $Z_1^\pm(x)$ from (6.4).

Theorem 7.3 (Convergent inverse sectors and the exact second sector)**.** *There are constants $c > 0$ and $w_0 > 0$ such that for each $w > w_0$ the three inverse displacements are holomorphic for $|q| < c/w$ when q is treated as an independent variable. At the physical substitution $q = \mathrm{e}^{-w}$, their convergent expansions give*

$$X(A(w)) = w + \frac{\log 2}{a_w} q^2 + O(w^3 q^3), \tag{7.6}$$

$$G_\pm(A(w)) = w - \frac{c_\pm}{a_w} q + \left(-\frac{b_w c_\pm^2}{2a_w^3} - \frac{c_\pm^2}{a_w^2} + \frac{c_\pm}{a_w} \right) q^2 + O(w^3 q^3). \tag{7.7}$$

The remainder constants can be chosen independently of large w. The physical branch in (7.6) is the global inverse of theorem 7.1.

Proof. For independent q, the defining equations for the displacements are

$$A(w + \delta) - A(w) - B(q\mathrm{e}^{-\delta}) = 0, \tag{7.8}$$

$$A(w + \delta) - A(w) + c_\pm \frac{q\mathrm{e}^{-\delta}}{1 + q\mathrm{e}^{-\delta}} = 0. \tag{7.9}$$

Fix a small radius $\rho > 0$, for example $\rho = 1/4$. The uniform estimates in theorem 6.4 imply

$$A(w + \delta) - A(w) = \frac{\delta}{w}(1 + O(w^{-1})) \qquad (|\delta| \le \rho).$$

For sufficiently large w, the core difference has exactly one zero in the disk and has boundary modulus at least $\rho/(2w)$; this follows by comparison with δ/w.

On the same disk, the lateral perturbation is bounded by $\pi|q|\mathrm{e}^\rho/(1 - |q|\mathrm{e}^\rho)$. Choose $c > 0$ small enough that this is less than $\rho/(2w)$ whenever $|q| \le c/w$, for all sufficiently large w. The physical perturbation $B(q\mathrm{e}^{-\delta})$ is $O(|q|^2)$ uniformly on the disk and satisfies the same strict inequality, after increasing w_0 if necessary. Rouché gives exactly one simple zero in each case. The implicit-function theorem and uniqueness give holomorphic roots on the parameter disk, bounded by ρ. Cauchy's estimate bounds their degree-n coefficients by $\rho(w/c)^n$. Hence on $|q| \le c/(2w)$ the remainder after degree two is at most

$$\rho \frac{(w|q|/c)^3}{1 - w|q|/c} = O(w^3|q|^3).$$

The physical value $q = \mathrm{e}^{-w}$ lies in this region for all large w. This proves actual sector convergence and a uniform remainder; it is not only a formal reversion computation.

For the coefficients, write $\delta = d_1 q + d_2 q^2 + O(q^3)$ and use

$$A(w + \delta) - A(w) = a_w \delta + \tfrac{1}{2} b_w \delta^2 + O(\delta^3), \qquad \frac{q\mathrm{e}^{-\delta}}{1 + q\mathrm{e}^{-\delta}} = q - (d_1 + 1)q^2 + O(q^3).$$

The first coefficient in the lateral equation gives $d_1 = -c/a_w$. The second gives $a_w d_2 + \frac{1}{2} b_w d_1^2 - c d_1 - c = 0$, which is precisely (7.7). In the physical equation, $B(q\mathrm{e}^{-\delta}) = (\log 2)q^2 + O(q^3)$, so $d_1 = 0$ and $d_2 = \log 2/a_w$. For the physical real q, conjugation and uniqueness make the physical root real. Its defining equation is $H(w + \delta) = A(w)$, and the global monotonicity of H identifies it with $X(A(w))$. □

7.4 A larger inverse error than the physical completion

Corollary 7.4 (Digamma-core arithmetic-averaging defect)**.** *With the preceding exact-core normalization,*

$$\frac{G_+(A(w)) + G_-(A(w))}{2} - X(A(w)) = \left(\frac{\pi^2}{a_w^2} + \frac{\pi^2 b_w}{2a_w^3} - \frac{\log 2}{a_w} \right) \mathrm{e}^{-2w} + O(w^3 \mathrm{e}^{-3w}) \tag{7.10}$$

$$\sim \pi^2 w^2 \mathrm{e}^{-2w}. \tag{7.11}$$

By contrast,

$$X(A(w)) - w \sim (\log 2) w \mathrm{e}^{-2w}, \qquad G_+(A(w)) - G_-(A(w)) \sim 2\pi \mathrm{i} w \mathrm{e}^{-w}. \tag{7.12}$$

In particular the inverse-average error divided by the physical core-inverse correction tends to infinity, with equivalent $\pi^2 w / \log 2$.

Proof. Average (7.7), using $c_+ + c_- = 0$ and $c_\pm^2 = -\pi^2$, then subtract (7.6). This gives (7.10). Since $a_w = w^{-1}(1 + O(w^{-2}))$ and $b_w = -w^{-2}(1 + O(w^{-2}))$, the coefficient of e^{-2w} in that formula is $\pi^2 w^2 + O(w)$. The uniform remainder is smaller by a factor $O(w\mathrm{e}^{-w})$. The same estimates give the first equivalent in (7.12). Subtracting the two lateral expansions gives its second equivalent. Division proves the ratio. □

This is a model-specific quantitative consequence of the general nonlinear averaging mechanism already proved in [9] and restated in [8]. What is added here is the explicit core, the exact second-sector coefficient, a uniform convergent chart, and the separation by an unbounded factor from the physical correction. It is *not* a counterexample to a correctly formulated nonlinear median summation. Nor does it allege that either cited repository article confused arithmetic averaging with such a summation.

In the target variable. Let $v = \mathrm{e}^{y-\gamma}$. Inverting (6.11) gives $w = v + \pi^2/(6v) + O(v^{-3})$. Therefore the physical inverse correction and the inverse-average error have the respective equivalents

$$(\log 2)\mathrm{e}^{y-\gamma} \exp(-2\mathrm{e}^{y-\gamma}), \qquad \pi^2 \mathrm{e}^{2(y-\gamma)} \exp(-2\mathrm{e}^{y-\gamma}). \tag{7.13}$$

These are doubly exponential scales in y. Using the exact w for finite computations remains preferable: an algebraic approximation to the core must not be confused with its much smaller exponential correction.

8 Verification, source audit, and integration safeguards

8.1 Retained finite evidence

The accompanying program `code/verify.py` runs without network access. The retained run used Python 3.13.5, SymPy 1.14.0, and mpmath 1.3.0 at 70 decimal digits. It passed **165 exact assertions** and **46 numerical assertions**, with no numerical group skipped. The exact checks include finite shift identities, operator inversions and ranks through degree ten, joint-root and Vandermonde checks through six pairs of nodes, normalized logarithmic-node filters through $N = 3$, and the second-order inverse coefficient calculation. Rational-node checks test universal polynomial algebra; actual logarithmic nodes are used separately where stated. None of these finite checks is a proof of the all-degree theorems.

For the Stieltjes formulas, the program extracts the coefficients of the *regularized* function $\zeta(1 - s, a) + 1/s$ by Cauchy sampling, at both 64 and 96 points, and compares the reconstructed first four physical jets at $x = 1.3, 3$ with the independent integral in theorem 5.2. This reference avoids the small-step polylogarithm-order differentiation issue recorded in [8]. Two tilted-ray Borel integrals are compared with the digamma formula. Additional checks cover three noninteger-order completion identities, six moving-order roots, five pairs of large inverse residuals, and three small-inverse residuals plus their first two coefficients. The largest absolute discrepancy among the 46 retained numerical assertions was below $2 \cdot 10^{-68}$. These are high-precision floating-point agreements, not interval bounds.

The asymptotic tables are diagnostics, not additional certified assertions about limits. Table 1 shows the normalized order detuning and the still-present completion on the neutral curve. Here $m = 1$, so the signed leading completion is $-q$.

The last displayed value is rounded; it does not mean exact equality. For the small-endpoint chart the error after six terms is about $3.90 \cdot 10^{-17}$ at $t = 10^{-2}$ and $3.88 \cdot 10^{-24}$ at $t = 10^{-3}$, consistent with a seventh-order remainder. Full values and residuals, rather than only these rounded summaries, are retained in `data/`.

Table 1: Moving-order diagnostics, rounded from the retained 70-digit run.

N	q	$\epsilon_N/(-\kappa_{1,N}q^N/\pi)$	$C/(-q)$
1	10^{-2}	0.9956011681	0.9946589499
1	10^{-3}	0.9995582649	0.9994621827
2	10^{-2}	0.9962146385	0.9877618322
2	10^{-3}	0.9996172905	0.9987664010
3	10^{-2}	0.9888879416	0.9895194578
3	10^{-3}	0.9988740742	0.9989479273

Table 2: Large-core inverse diagnostics. The last column compares the actual inverse-average error with the full second-sector term in (7.10), not just its leading equivalent.

w	$(X-w)/(we^{-2w})$	Defect$/(w^2e^{-2w})$	Defect/second sector
5	0.5853514479	5.9934294047	0.9633235152
8	0.6485114953	7.9349324606	0.9988202952
12	0.6756899051	8.9089971386	0.9999811029
20	0.6872810572	9.4220413126	0.9999999938
35	0.6912692028	9.6554457036	1.0000000000
Limit	$\log 2$	π^2	1

8.2 Corrections and qualifications that belong with this package

No newly verified mathematical error is attributed to the inspected repository sources. The following are precise integration safeguards, not invented corrections to unseen articles.

Do not discard a zero-times-pole product at order zero. The factor $1/\Gamma(s)$ vanishes at $s = 0$, but the Hurwitz zeta factor in (2.4) has a pole there. Their product has the nonzero value $L_0^{\pm} = 1$. Evaluating the factors separately would lose all the Stieltjes-jet information.

Do not identify zero jump with physical reconstruction. The exact neutral curve in theorem 4.4 has zero jump and a nonzero completion starting at action one. Conversely, cancelling both fixed-order data costs twice as many prescribed roots, as theorem 3.2 proves. Neither assertion by itself removes all singularities on all continued Borel sheets.

Keep differentiation and continuation hypotheses. The boundary integral for the normalized lattice kernel was proved in $\operatorname{Re} s > 0$. At zero we used analytic continuation and then a separately proved meromorphic kernel. A bare substitution into a nonintegrable boundary power is not an alternative proof. Order derivatives also act on $x^s/\Gamma(s+1)$, not just on an isolated Borel minor.

Keep the exact inverse core and the meaning of averaging. The physical correction to $w = A^{-1}(y)$ is exponentially small. Replacing w by a low-order algebraic approximation generally creates a larger error. The arithmetic average of lateral inverses differs from the physical inverse by the explicit term (7.10); no claim is made here about constructing an algebra-compatible median summation.

Do not promote a sufficient disk to an exact radius. The small-target radius $\sqrt{3}/40$ and the large-core parameter radius c/w are proved sufficient. They are not claimed optimal. Likewise, real monotonicity of H does not prove complex univalence of its global continuation.

8.3 Proposed integration and claim boundary

The package should be placed in a new directory such as

```
Analysis/Transseries/docs/series-and-transseries/
Polylogarithm_Order_Jet_Filters/
```

with a backlink from the polylogarithm reports. The companion `integration/INTEGRATION.md` identifies proposed insertion points and result labels. The existing resonant-polylogarithm package should remain the credited source of the positive-integer Gamma/Bell formulas, and the existing transport package should remain the credited source of the general inverse mechanism.

The proofs in this article are complete within their stated hypotheses. The source comparison is bounded, and the proposed novelty of the assembled results has not been independently refereed or cleared against every incoming draft. The package contains neither a Lean development nor an interval-certified numerical implementation. Its reproducible finite checks supplement, rather than replace, the mathematical proofs.

9 Further research questions

1. Global Borel normal forms for higher order jets. The exact dictionary controls the entire exponential jump, but not an explicit normal form for every local singular germ of a higher-jet Borel transform. Develop those germs in a form compatible with the Stieltjes coordinates and determine which fixed-order filters remove a singularity, which only suppress a jump coefficient, and what occurs on later sheets. This continues Research Question 4 of [8]; it is not resolved by the finite root-counting theorem alone.

2. Conditioning and optimal normalization of deep filters. The minimal degrees are exact, but the coefficients of the extremizing polynomials can be large. Determine norms and condition numbers for interpolation on $-\log n \pm i\pi$, and optimize cancellation under a fixed coefficient norm or a restriction to a prescribed coefficient field. Exact degree optimality is not numerical optimality.

3. A uniform theory with growing cancellation depth. The neutral-curve theorem fixes N before $q \to 0$. Determine the largest ranges $N = N(x) \to \infty$ for which the crossover and the implicit curve have uniform errors, and the competing growth of Π_N, $\kappa_{m,N}$, and the convergence radius. The individual fixed-N proofs do not justify this joint limit.

4. Neutral curves under nonlinear inverse transport. When the order is selected as a function of $q = e^{-x}$, inversion must include the induced order dependence in its derivative. Construct inverse charts along the exact neutral curve, and determine whether jump cancellation remains useful after the physical completion and this additional chain rule are both retained. A forward identity alone does not solve that nonlinear problem.

5. The exact small-target inverse radius of H. Locate the closest complex critical value or singular value of $z\sqrt{B(z)/(\log 2\, z^2)}$. Is the dominant obstruction a simple fold, the logarithmic boundary at $z = -1$, or another continued singularity? This would replace the explicit sufficient disk in theorem 7.2 by a sharp theorem, with coefficient asymptotics rather than only local convergence.

6. Higher zero-order jets and their inverse scales. The leading algebraic terms of higher jets are Bell polynomials in $\log x + \gamma$ and zeta values. Classify their eventual monotonicity, critical points, and inverse branches, and determine which parity-induced completion actions become doubly exponential sectors after inversion. The global positivity argument here is special to the first jet and cannot simply be asserted for the higher logarithmic moments.

7. Combining periodic weights and order filters. The periodic-weight dictionary in [8] can be combined with order differentiation. Determine optimal designs when both the period and jet degree are constrained, keeping the zero Fourier mode. In particular, quantify the tradeoff between removing residue classes and removing individual logarithmic spectral nodes; the two forms of arithmetic freedom should not be conflated.

8. Formal proofs and certified numerical continuation. The principal-ideal theorem and finite operator inversions are practical first formalization targets over real polynomials. The transcendence corollary can import Gelfond–Schneider rather than recreate it. For the analytic part, certify the regularized integrals, tilted Borel contours, and inverse disks with interval enclosures and explicit tail bounds. The existing numerical scripts deliberately do not claim these certificates.

A Reproduction and theorem map

The source is self-contained and uses standard TeX packages; no external images or custom font files are required. From the package root:

```
python -m pip install -r requirements.txt
python code/verify.py --output-dir replay
make pdf
```

A single verification group can be rerun with, for example,

```
python code/verify.py --group stieltjes --output-dir replay-stieltjes
```

The program refuses to overwrite existing output unless `-overwrite` is explicitly provided. Numerical precision is selectable with `-dps`, with a minimum of 50 digits. The retained run uses 70. Each group records its versions, precision, finite assertion counts, and numerical differences. The supplied build-validation record reports the actual PDF checks; it should not be confused with a formal mathematical certificate.

Result	Status	Main dependency
Theorem 2.1	Credited background	Classical Fermi split, Jonquière identity; explicit repository realization.
Theorem 2.3	Derived exact dictionary	Finite order operator and normally convergent tail.
Theorems 3.1 and 3.2	Proposed sharp classification	Finite differential operators, polynomial roots, conjugate interpolation.
Theorem 4.2	Derived all-order rigidity	Gelfond–Schneider, imported as a classical theorem.
Theorem 4.4	Proposed analytic cancellation theorem	Sharp normalized filter and holomorphic implicit functions.
Theorems 5.1 and 5.2	Exact order-zero synthesis	Hurwitz Laurent expansion, reciprocal Gamma, endpoint subtraction.
Theorem 6.3	Proved zero-order kernel extension	Digamma integral and a specified contour rotation.
Theorems 7.1 and 7.2	Proposed endpoint package	Positive integral, convergent series, Lagrange inversion, Rouché.
Theorems 7.3 and 7.4	Model-specific transport	Exact digamma core; known general inverse architecture credited.

References

[1] NIST Digital Library of Mathematical Functions, *§25.12, Polylogarithms*, especially Eqs. 25.12.11, 25.12.13, and 25.12.16. https://dlmf.nist.gov/25.12. Accessed 12 October 2026.

[2] NIST Digital Library of Mathematical Functions, *§5.5, Functional Relations; §5.7, Series Expansions*. https://dlmf.nist.gov/5.5; https://dlmf.nist.gov/5.7. Accessed 12 October 2026.

[3] NIST Digital Library of Mathematical Functions, *§5.9, Integral Representations*, digamma integrals. https://dlmf.nist.gov/5.9. Accessed 12 October 2026.

[4] NIST Digital Library of Mathematical Functions, *§5.11, Asymptotic Expansions*, especially Eq. 5.11.2. https://dlmf.nist.gov/5.11. Accessed 12 October 2026.

[5] NIST Digital Library of Mathematical Functions, *§25.11, Hurwitz Zeta Function*. https://dlmf.nist.gov/25.11. Accessed 12 October 2026.

[6] A. Gil, J. Segura, and N. M. Temme, Complete asymptotic expansions for the relativistic Fermi–Dirac integral, *Applied Mathematics and Computation* **412** (2022), 126618. https://doi.org/10.1016/j.amc.2021.126618. Inspected preprint: arXiv:2108.11210v2, Section 3 and Lemma 3.1, https://arxiv.org/html/2108.11210v2.

[7] M. Karatarakis and F. Wiedijk, *A formalization of the Gelfond–Schneider theorem*, arXiv:2603.24823v1 (25 March 2026), especially Theorem 3.1 and its complex-logarithm convention. https://arxiv.org/html/2603.24823v1. The classical theorem is imported here; no formalization of the present article is claimed.

[8] ProveIt research package, *Resonant Polylogarithm Transseries: Gamma Borel Kernels, Cyclotomic Order Jets, and Sharp Dilogarithm Reversion*, 11 October 2026. `Analysis/Transseries/docs/series-and-transseries/Resonant_Polylogarithm_Transseries/article.tex`. Snapshot `8f5afd4d7565df808359f6a0c22f87f79a119bb9`; source blob `d54638582ab147eb4a48a846c3bdc371643aaf39`. Theorems identified in the source by labels `thm:lattice`, `thm:Gamma-kernel`, `thm:Bell`, and `thm:periodic`; inversion and audit sections were also inspected as recorded in the accompanying source audit.

[9] ProveIt research package, *Nonlinear Stokes Transport under Logarithmic Inversion*, `Analysis/Transseries/docs/series-and-transseries/Nonlinear_Stokes_Transport_Logarithmic_Inversion/article.tex`. Snapshot `8f5afd4d7565df808359f6a0c22f87f79a119bb9`; source blob `1e8b24325554d1f1c94c29f21b794ba62feff755`. The general contour inverse coefficients and the explicit distinction between arithmetic averaging and nonlinear-compatible median summation are prior repository material.

[10] ProveIt Contributors, *Polylogarithms project README; Series and transseries inventory; Incoming-directory listing*, inspected at snapshot `8f5afd4d7565df808359f6a0c22f87f79a119bb9`. `Analysis/Polylogarithms/README.md`; `Analysis/Transseries/docs/series-and-transseries/README.md`; `docs/incoming/`. Repository: https://github.com/VladimirReshetnikov/ProveIt. The listing alone does not establish the contents of its binary archives.